26

Singularities and Quantum Gravity

In *Interstellar* Cooper and TARS seek quantum data inside Gargantua, data that could help the Professor solve his equation and lift humanity off Earth. The data, they believe, must reside inside a singularity that inhabits Gargantua's core—a "gentle" singularity, Romilly predicts. What are the quantum data? How could they help the Professor? And what is a gentle singularity?

The Primacy of Quantum Laws



Our universe is fundamentally quantum. By this I mean that everything fluctuates randomly, at least a little bit. Everything!

When we use high-precision instruments to look at tiny things, we see big fluctuations. The location of an electron inside an atom fluctuates so rapidly and so randomly, that we can't know where the electron is at any moment of time. The fluctuations are as big as the